

SPRITE

Tier 1 :: Data Fabrication

At a glance: SPRITE (Sample Parameter Reconstruction via Iterative Techniques) goes beyond asking whether a single reported mean is arithmetically possible. Given a reported mean, standard deviation, sample size, and response scale, SPRITE reconstructs the full space of integer datasets that could have produced those summary statistics. When no such dataset exists, or when every possible dataset implies physically implausible patterns — such as responses clustered impossibly at one end of a scale — the paper contains a fundamental inconsistency. Like GRIM, SPRITE requires only published summary statistics and no raw data.

1. Overview

The GRIM test answers a binary question: is this reported mean arithmetically possible given the sample size and integer scale? SPRITE asks a richer question: what do the actual response distributions look like that could have produced this combination of mean, standard deviation, and sample size? Rather than a pass/fail verdict, SPRITE returns a picture — a set of plausible datasets — and asks whether that picture is coherent with reality.

SPRITE was developed by James Heathers, Jordan Anaya, Tim van der Zee, and Nick Brown, and first circulated as a preprint in 2018. It extends the logic of GRIM in two important directions. First, it incorporates the standard deviation as well as the mean, which dramatically tightens the constraints on what datasets are possible: a mean that is achievable by many integer distributions may be achievable by only a handful when the SD is also constrained. Second, it can reveal that even though a mean-SD combination is technically possible, every dataset that could produce it is physically implausible — implying extreme distributions that no realistic sample of participants would generate.

The method has its most powerful application in studies using bounded integer scales — Likert-type rating instruments, binary outcomes, small-range counts — where the constraints on possible distributions are strongest. It is less informative for continuous measurements or very large samples, where the space of possible distributions becomes so large that the constraints lose their bite.

2. Methodological Overview

SPRITE works by computational enumeration. The algorithm generates a large number of random integer datasets of size N drawn from the reported scale range $[min, max]$, then iteratively adjusts pairs of values until both the target mean and target standard deviation are simultaneously achieved. Each valid dataset found is a member of the SPRITE solution set — a plausible reconstruction of what the underlying data could have looked like.

The iterative reconstruction algorithm

Starting from a random distribution with the correct mean (obtained by adjusting values until $\Sigma x/N$ matches the reported M), the algorithm applies the following iterative step:

1. Select two values at random: x_i and x_j from the current distribution.

The swap step: move x_i and x_j slightly toward or away from each other by 1 unit.

2. Accept the swap if it brings the current SD closer to the target SD.

Reject it if the swap moves the SD further away, or if it pushes x_i or x_j outside $[min, max]$.

3. Repeat until SD matches the target (within rounding tolerance).

Each successful run produces one member of the solution set. Run thousands of times to explore the full solution space.

The result is a large sample from the space of possible datasets. This sample can then be examined visually (as a histogram grid showing each reconstructed distribution) or analytically (checking whether the distributions cluster in plausible regions of the scale or exhibit impossible features such as bimodality with all responses at the extremes).

What SPRITE detects

Three types of anomaly are identifiable from the SPRITE solution set:

- Empty solution set: no integer dataset of size N can simultaneously achieve the reported M and SD within rounding tolerance. This is a direct impossibility — the summary statistics are internally inconsistent, regardless of what the underlying data looked like. This is a stronger finding than a GRIM failure alone.
- Implausible distribution shapes: the solution set exists but every distribution in it has an extreme or physically implausible shape. A classic example is a distribution where all N participants must have responded at either the minimum or maximum of the scale, with no intermediate responses — a pattern inconsistent with normal human response behavior on a rating scale. Or a distribution where more than half of participants gave identical responses, when the study design makes such clustering highly improbable.
- Inconsistency across subgroups: a paper reports means and SDs for multiple subgroups, each of which passes SPRITE individually. But when the subgroup distributions are combined, the resulting pooled distribution is impossible given the reported overall mean or SD. SPRITE can detect this cross-subgroup inconsistency by simultaneously constraining multiple distributions.

3. SPRITE vs. GRIM: Complementary Constraints

GRIM and SPRITE operate on the same class of data — integer-scale responses with reported summary statistics — but they constrain different aspects of the summary and therefore catch different classes of error. Understanding the relationship helps clarify when each is more useful.

What it tests	GRIM: Is the reported mean achievable for any integer dataset of this N? SPRITE: What distributions could produce this exact M and SD combination?
Information used	GRIM: Mean and N only. SPRITE: Mean, SD, N, scale minimum, and scale maximum — all five.
Output	GRIM: Binary pass/fail verdict. SPRITE: A set of plausible distributions, which may be empty, implausible, or normal.
Error types detected	GRIM: Impossible means (fabricated or transcription error). SPRITE: Impossible M-SD combinations; implausible distributions; cross-subgroup inconsistencies.
Power vs. N	GRIM: Loses power as N grows (many decimal means become achievable). SPRITE: Also weakens at large N, but SD constraint keeps it informative for moderate N (up to ~100).
Computational cost	GRIM: Instant — single arithmetic check. SPRITE: Requires iterative search; computationally heavier but still fast (seconds per statistic in R).

In practice, SPRITE is applied after GRIM as a second-pass analysis. GRIM rules out impossible means quickly; SPRITE then interrogates the plausibility of the underlying distribution for the statistics that pass the GRIM check but still merit scrutiny.

4. Example Application: Wansink Food Lab

The most extensively documented application of SPRITE is to Brian Wansink's Cornell Food and Brand Lab papers, the same set that prompted the original GRIM analysis by Brown and Heathers. After GRIM identified dozens of impossible means across the lab's published papers, Heathers and colleagues applied SPRITE to the mean-SD pairs that had survived the GRIM screen.

One set of findings involved a study on how plate size affects the amount of food served. The paper reported that children given large plates served themselves a mean of 59.8 grams of pasta with an SD of 47.3, N = 43. GRIM passes this ($59.8 \times 43 = 2571.4$ — not quite an integer, but within rounding tolerance for one decimal place). SPRITE, however, revealed that every distribution capable of producing $M = 59.8$, $SD = 47.3$, $N = 43$ on a plausible gram-range scale involved an implausible concentration of values at very low quantities (near zero grams) combined with a few extreme outliers at very high quantities. The shape was inconsistent with a normal serving-size distribution from a real cafeteria study.

A second example: a study reported a mean of 2.50, SD = 0.54, N = 20 on a 5-point scale. SPRITE found that the only distributions satisfying these constraints had exactly 20 participants responding exclusively with 2s and 3s — a distribution so perfectly symmetric and limited in range that it is extraordinarily unlikely to occur in real behavioral data, where participants typically show more variability.

These analyses, combined with the GRIM findings, contributed to the broader pattern of evidence that led Cornell University to investigate the lab. The investigation ultimately concluded that data had been manipulated, and 13 papers were retracted.

5. Requirements & Applicability

SPRITE shares the core requirement of GRIM — integer-scale response data with reported summary statistics — but adds the standard deviation as an essential input. This means it is applicable to a narrower set of reported statistics than GRIM, but each application is more powerful.

Mean (M)	The reported mean to four significant figures. More decimal places provide stronger constraints. A mean rounded to one decimal place leaves a wider solution window than a mean reported to two or three.
Standard deviation (SD)	Required — this is what distinguishes SPRITE from GRIM. The SD must be reported with at least one decimal place. If only standard errors (SE) are reported, these must be converted: $SD = SE \times \sqrt{N}$.
Sample size (N)	Required. SPRITE's constraints tighten as N decreases; the method is most powerful for $N < 100$, and the solution space becomes very large for N above ~200.
Scale minimum and maximum	Required. SPRITE generates integer values within [min, max]. An incorrect scale range (e.g., treating a 0–6 scale as a 1–7 scale) will generate a misleading solution set. Scale bounds must be verified from the paper's methods section.
Integer scale	The underlying data must be integer-valued. Continuous measurements, log-transformed variables, or composite scores computed from multiple items cannot be tested directly. Individual item responses on a Likert scale are SPRITE-applicable; scale means aggregated across items are not (their values are not integers).
Raw data required?	No. Like GRIM, SPRITE operates entirely on published summary statistics. This is the key advantage: it can be applied by any reader with access to the paper, without requiring data sharing from the original authors.
Number of statistics	SPRITE is typically applied to a set of statistics from the same paper, not just a single M-SD pair. Patterns of anomaly across multiple statistics in the same paper are more diagnostic than any individual result.

6. Limitations

- SPRITE loses power rapidly as N increases. For N above ~200, the space of possible integer distributions is so large that almost any M-SD combination is achievable, and the solution set rarely has a constrained shape. The method is primarily useful for small-to-medium sample studies ($N < 100$).
- The method requires correct scale bounds. If an analyst uses the wrong scale range — for example, treating a 7-point scale anchored at 0 as running from 1 to 7 — the solution set will be miscalculated. Always verify scale range from the methods section before running SPRITE.
- SPRITE identifies implausible distributions, but implausibility is a judgment, not a bright line. A solution set containing only extreme bimodal distributions is highly suspicious; a solution set with mild skew is less so. Interpreting SPRITE output requires domain knowledge about what response distributions look like for the specific construct and population being studied.
- Rounding cascades can produce false positives. If means and SDs were both rounded independently before reporting, the combination may appear to violate SPRITE constraints even when the underlying data were entirely legitimate. More decimal places in reported statistics reduce this risk. When a SPRITE solution set is empty by a tiny margin, rounding error is a plausible explanation.
- SPRITE cannot determine the cause of an anomaly. An empty solution set or implausible distribution could reflect fabrication, data manipulation, copy-paste errors, incorrect scale reporting, or a genuinely

unusual data distribution. SPRITE flags the inconsistency; establishing its cause requires additional investigation.

- The iterative algorithm is stochastic. Different runs may produce slightly different solution sets, especially for distributions near the boundary of feasibility. Running SPRITE multiple times (the rSPRITE R implementation handles this automatically) and examining the aggregate solution space is better practice than relying on a single run.

7. How to Apply

rSPRITE (R package)

The rSPRITE package (Brown & Heathers, maintained on GitHub) is the standard implementation. It accepts a data frame of (mean, sd, n, scale_min, scale_max) rows and for each row generates a set of reconstructed distributions, plots them as histograms, and flags cases where the solution set is empty or all solutions are implausible.

Minimal workflow

- Install the package: `devtools::install_github('nicebread/rSPRITE')` in R.
- Build the input: create a data frame with columns mean, sd, n, min_val, max_val for each M-SD pair to test. Verify scale bounds from the paper's methods section before entering them.
- Run the analysis: `sprite_results <- rsprite2(df)`. The function returns a list of solution sets, one per row. For each, inspect the histogram grid of reconstructed distributions.
- Flag anomalies: cases where `solution_count == 0` are impossible. Cases where all solutions show extreme concentration at scale endpoints or implausibly low within-distribution variance warrant documentation and further scrutiny.
- Apply across multiple statistics: SPRITE is most informative when run on all reported M-SD pairs in a paper simultaneously. Patterns of anomaly — multiple consecutive failures, or all solutions in multiple analyses being implausible — provide stronger evidence than any individual result.

Online tool

A web implementation is available at the SPRITE online app maintained by Brown and Heathers (searchable via GitHub). It accepts individual M, SD, N, min, max entries and returns a visual plot of the solution space without requiring R installation — useful for quick screening of individual statistics before committing to a full batch analysis.

8. Related Tools

GRIM Test	The natural prerequisite: GRIM checks whether the reported mean is possible for any integer dataset before SPRITE asks what the distribution looked like. Always run GRIM first. A GRIM failure is sufficient to flag the statistic; SPRITE adds value when the mean passes GRIM but the M-SD combination still seems anomalous.
GRIMMER	Extends GRIM to standard deviations: checks whether the reported SD is achievable given the reported M, N, and integer constraints, using a more complex calculation. GRIMMER and SPRITE address the same problem from different angles — GRIMMER asks whether SD is possible; SPRITE asks what distribution could produce the M-SD pair together. GRIMMER is faster; SPRITE is more informative.
statcheck	Automated extraction and recalculation of p-values. Operates on test statistics rather than means and SDs; catches a different class of reporting error. Complementary to SPRITE in that both can be applied post-publication using only the published paper.
Multiverse Analysis	If raw data are available, multiverse analysis reveals whether results are robust across defensible analytic choices. SPRITE and GRIM are the post-publication counterparts for integer-scale data: they apply when raw data are unavailable and only summary statistics are reported.

Key Citations

- Heathers, J. A. J., Anaya, J., van der Zee, T., & Brown, N. J. L. (2018). Recovering data from summary statistics: Sample parameter reconstruction via iterative techniques (SPRITE). *PeerJ Preprints*, 6, e26968v1. <https://doi.org/10.7287/peerj.preprints.26968v1>
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